



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -2

CO1_AT2

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Course code : CSA1613

Course name: Data Warehousing And Data Mining

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TECHNICAL PRESENTATION



1. Introduction

Dimensional Data Modeling is a data warehouse design technique used to organize large amounts of data for analysis and reporting. It makes it easier for users to retrieve information quickly and supports business intelligence applications. This model is widely used in industries such as banking, healthcare, retail, and e-commerce to analyze historical data and improve decision-making.

2. Components of Dimensional Data Modeling

2.1 Fact Table

A Fact Table stores measurable or numerical data related to business activities. It contains foreign keys that connect to dimension tables.

Examples:

- Sales Amount
- Transaction Count
- Profit

- Quantity Sold

Features

- Stores numerical values.
- Contains foreign keys.
- Used for business analysis.

2.2 Dimension Table

A Dimension Table stores descriptive information about the facts. It provides details that help users understand the data.

Examples:

- Customer Name
- Product Name
- Branch Name

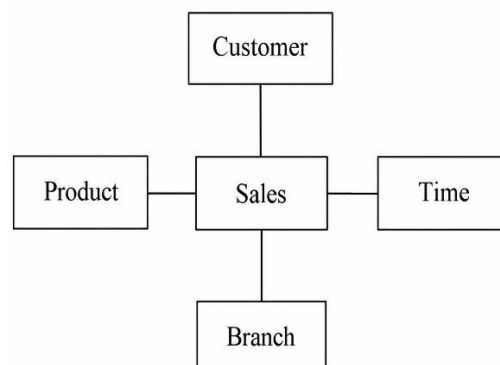
Features

- Contains descriptive attributes.
- Helps filter and group data.
- Connected to the Fact Table.

3. Star Schema

A Star Schema is the simplest dimensional model. It has one Fact Table connected directly to multiple Dimension Tables, forming a star-like structure.

Diagram



Features

- Simple and easy to understand.
- Faster query performance.
- Uses more storage space.

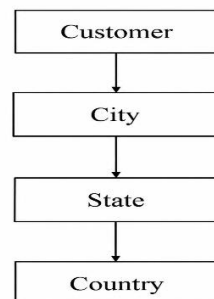
Advantages

- Easy to design.
- Fast data retrieval.
- Best for reporting and analysis.

4. Snowflake Schema

A Snowflake Schema is an extension of the Star Schema. Here, dimension tables are divided into smaller normalized tables to reduce data redundancy.

Diagram



Features

- Reduces data redundancy.
- Uses less storage.
- More complex than Star Schema.

Advantages

- Better data integrity.
- Saves storage space.

5. Real-World Example

Banking System

A bank stores transaction data in a Fact Table and customer, branch, account, and time information in Dimension Tables.

Using Dimensional Data Modeling, the bank can answer questions such as:

- Which branch has the highest number of transactions?
- Which customer performs the most transactions?
- What are the monthly transaction trends?
- Which account type is used the most?

6. Conclusion

Dimensional Data Modeling is an important technique used in data warehousing to organize data for easy analysis and reporting. The Star Schema is simple and provides faster query performance, while the Snowflake Schema reduces data redundancy and saves storage space. Both schemas help organizations analyze data efficiently and support better business decision-making.